



UITK VirtuaStick – User Guide

Thank you for choosing **UITK VirtuaStick**, a fully customizable and extendable virtual joystick built with **UI Toolkit (UITK)** and compatible with **Unity's Input System**.

For support, feedback, or feature requests, feel free to contact:

 empress.games.20@gmail.com



Table of Contents



[UITK VirtuaStick – User Guide](#)



[Table of Contents](#)



[Getting Started](#)

[Importing the Package](#)

[Accessing the Demo Scenes](#)



[Quick Integration Guide](#)



[Demo Scenes Overview](#)

[Core Features & Configuration](#)

[USS Styles](#)

[Character Control](#)



[Example Showcase](#)



[Additional Notes](#)



[Roadmap \(Planned Features\)](#)



[Version](#)



Getting Started

Importing the Package

Open Unity Package Manager and navigate to:

`Window > Package Manager > UITK VirtuaStick > Samples`

Accessing the Demo Scenes

Scenes are located at:

`Assets/Samples/UITK VirtuaStick/[Version]/`



Quick Integration Guide

To integrate VirtuaStick into your project:

1. Add the joystick via UXML or attach the `VirtuaStickProcedural` component to a `UIDocument`.
2. Add the `OnScreenVirtuaStick` component to the same `GameObject`.
3. Assign the correct binding path for your Input Action.
4. In your character or camera script, read and apply the action using Unity's Input System callbacks.

This modular setup allows VirtuaStick to adapt easily to any gameplay system.



Demo Scenes Overview

Core Features & Configuration

Test and tweak VirtuaStick functionality in real time.

Key Features:

- Dynamic mode toggle
- Axis restriction (Free / Horizontal / Vertical)
- Adjustable movement radius
- Real-time debug overlay

Use this demo to explore how VirtuaStick responds to various runtime settings and inputs.

USS Styles

Explore how Unity Style Sheets (USS) can customize the look and feel of VirtuaStick.

Evaluate:

- Background and handle visuals
- Borders and outlines
- Theming for light/dark UI modes

Refer to [USS_Styles.uxml](#) to adapt and extend styles for your own UI themes.

Character Control

Demonstrates a mobile-style dual-stick configuration:

- Left stick: character movement
- Right stick: camera rotation

Integration Summary:

- Add each stick via UXML or [VirtuaStickProcedural](#)
- Attach [OnScreenVirtuaStick](#) to each
- Assign correct binding paths
- Apply Input Actions in your controller scripts (player and camera)

Includes UI buttons to:

- Enable/disable dynamic mode
- Restrict axis movement
- Show/hide debug visuals

This is your go-to reference for gameplay integration.

Example Showcase

Presents four specialized extensions of VirtuaStick for different genres:

Variant	Description
<code>VirtuaStickDpad</code>	Standard joystick with D-Pad visual overlay
<code>VirtuaStickPuzzleDpad</code>	Orthogonal-only movement (no diagonals), minimal handle
<code>VirtuaStick2DPlatformer</code>	Horizontal-only movement with visual arrows (ideal for platformers)
<code>VirtuaStickRacingPad</code>	Steering-style circular input with dynamic rotation (no visible handle)

Each is implemented as a subclass of `VirtuaStick` and located in the same folder as the scene for easy reference and extension.

Use this demo as inspiration for customizing joystick behavior per genre.

Additional Notes

-  Requires Unity Input System v1.4+
 -  Verified with:
 - Unity LTS: 2021.3.45f1
 - Unity Tech Stream: 6000.1.15f1 (UITK stable release)
 -  Not compatible with the legacy Input Manager.
-



Roadmap (Planned Features)

- Built-in D-Pad visualization system
 - Editor tooling for configuration and preview
 - Additional USS style presets
 - Multitouch gesture support
-



Version

This documentation reflects **UITK VirtuaStick v1.0.0**.
Stay up to date via the [Unity Asset Store listing](#).